Battle of the Chinese Farm

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Agenda

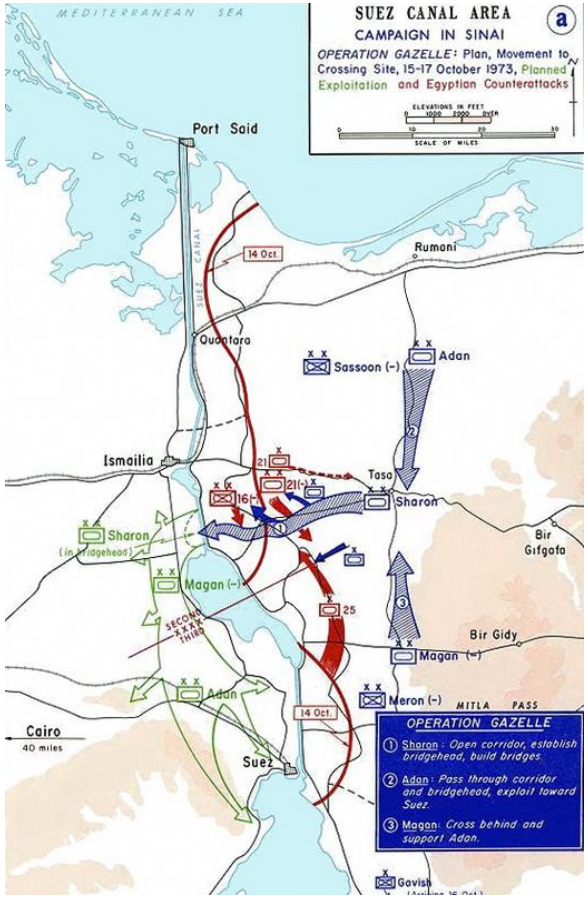
1. Introduction & Research Question
2. Battle Overview & Key Issues
3. Behavior Models
4. Simulation Setup & Parameters
5. Results & Analysis
6. Strategic Implications

Battle of the Chinese Farm

October 15-17, 1973 Arab-Israeli War (Yom Kippur War)

- Israeli and Egyptian forces in Sinai Peninsula
- Heavy casualties on both sides
- Israeli victory in battle

Key Question: Would the Egyptian army have been able to hold on to their territorial advantage if they took an offensive strategy against the Israeli bridgehead?



"Battle of the Chinese Farm." *Wikipedia*, Wikimedia Foundation, https://en.wikipedia.org/wiki/Battle_of_the_Chinese_Farm. Accessed 23 Jan. 2025.

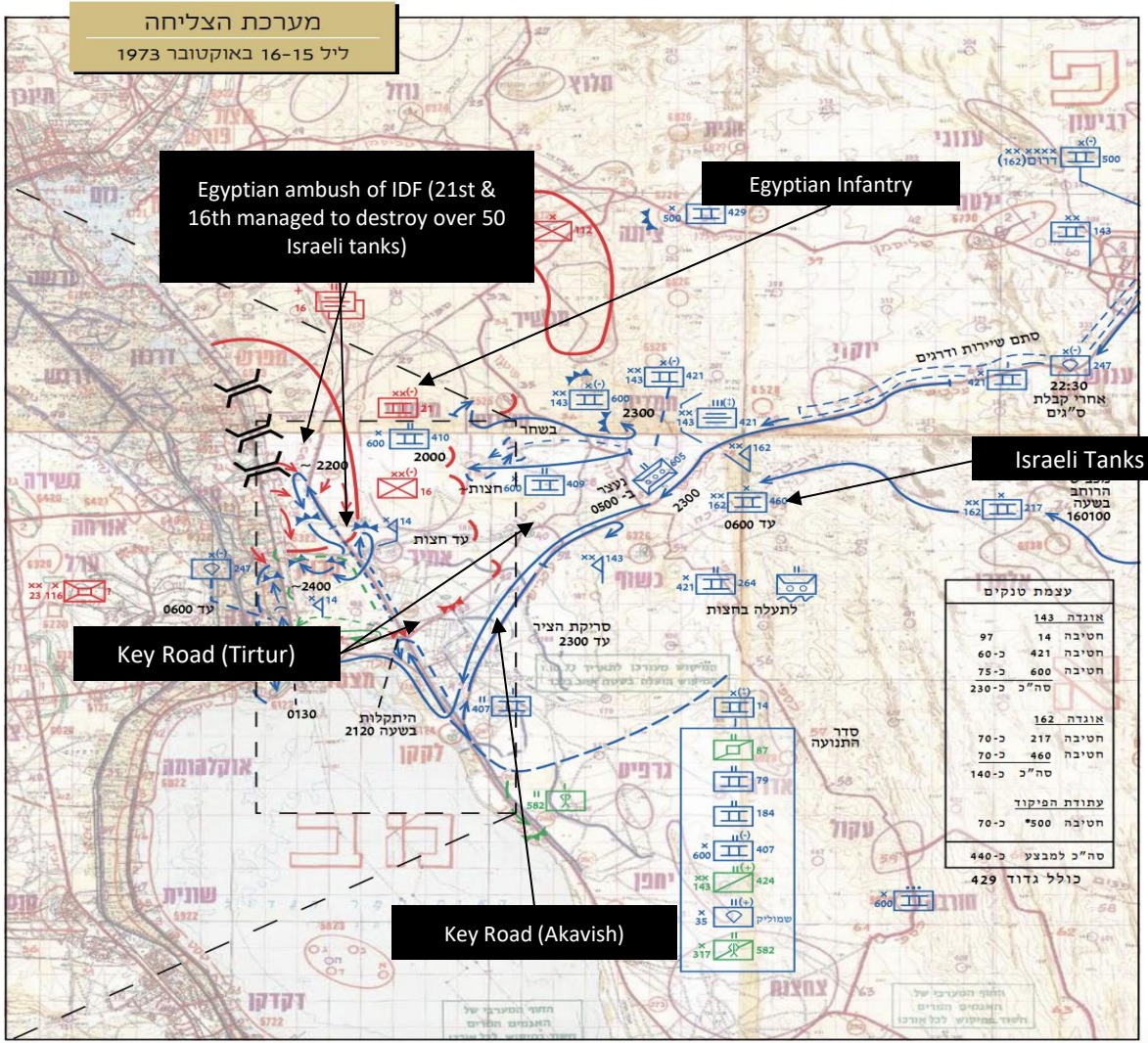
Research Question & Hypothesis

RQ: *How did the Egyptian army's rigidity in risk-averse positioning impact territorial control against the Israeli offensives in the Battle of the Chinese Farm?*
Independent Variable: Positioning Strategy
Dependent Variable: Territory Control

Hypothesis: The Egyptian army's focus on a defensive strategy prevented adjustments to defense priorities in response to the offense of the Israeli bridgehead. It is expected that by increasing these abilities, the Egyptians should maintain their positions and keep their advantage.

Progression of Battle

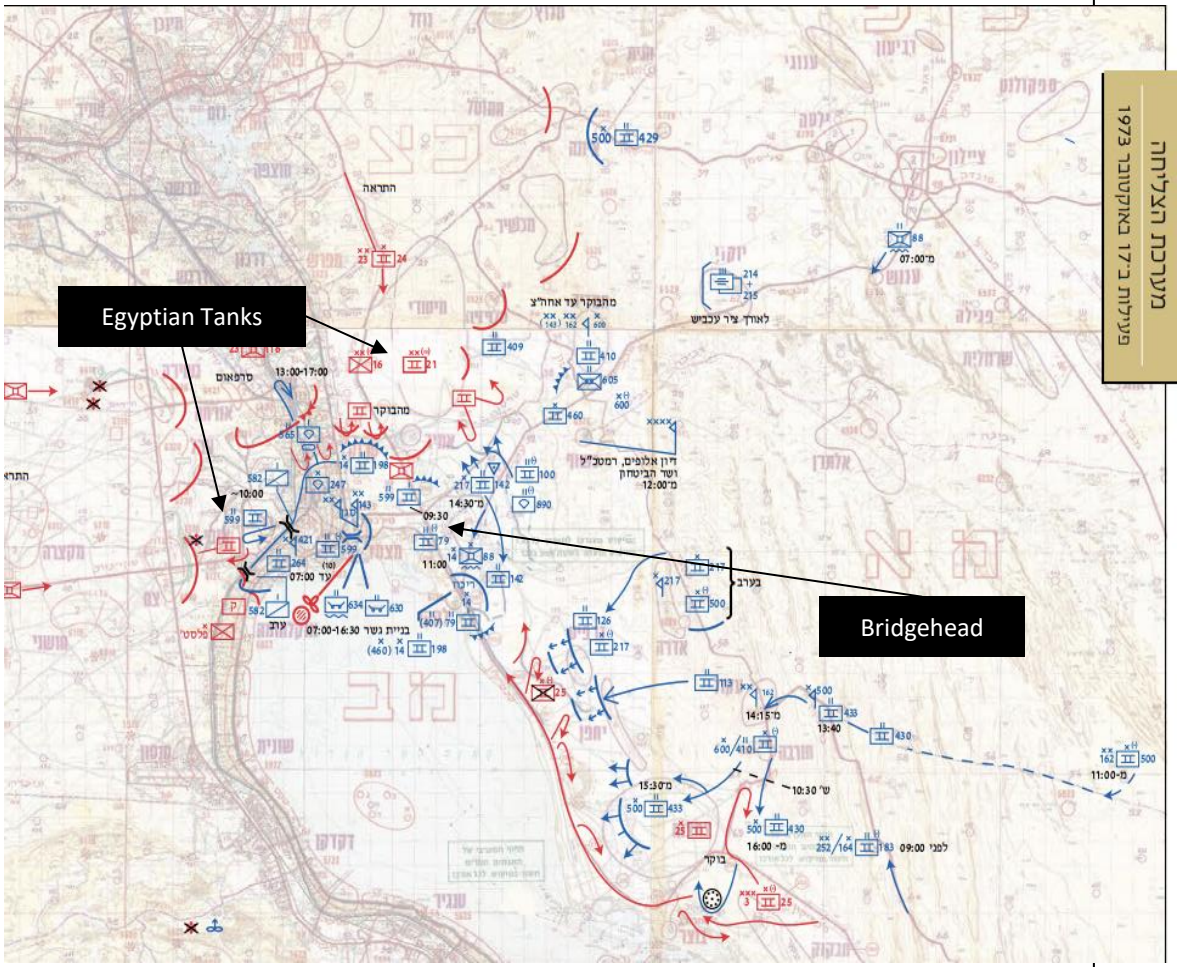
- Israeli & Egyptian Goals
- Egyptian advantage in beginning
- Israelis expand bridgehead under heavy fire
- Impactful errors on both sides



Egyptian Behavior

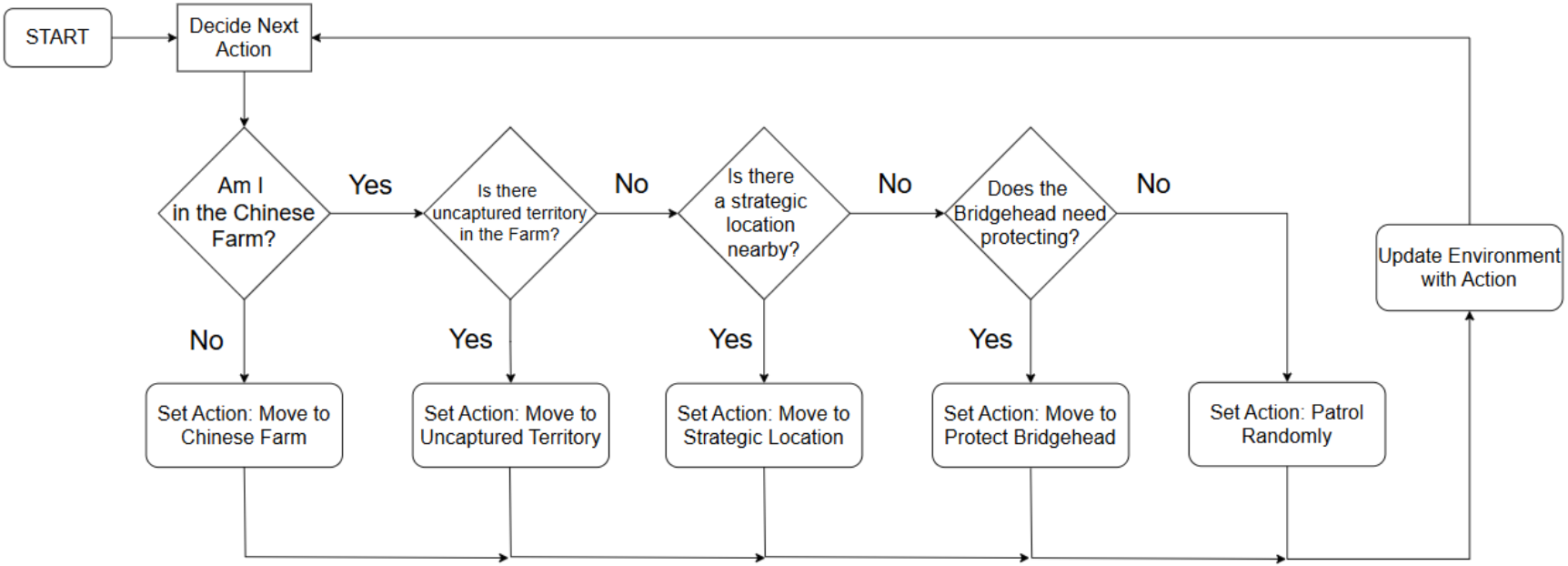
- **Risk-Averse Positioning:**
  - Small counterattacks
  - Static defense
- **Movement of Troops:**
  - Defensive positions instead of moving troops to bridgehead on west bank

End of Battle: Israeli victory



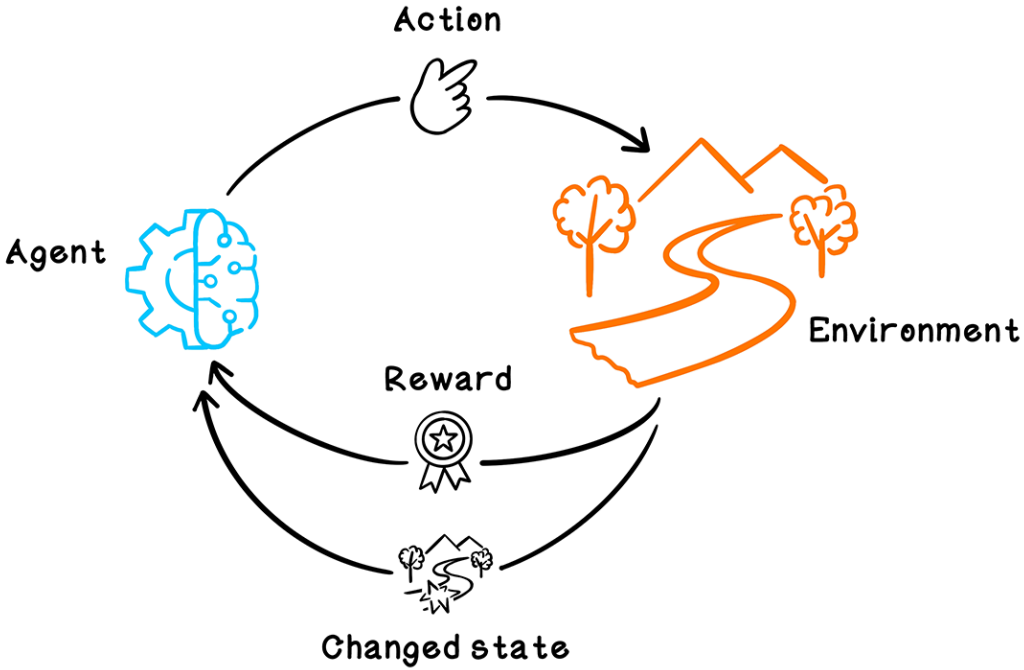
Behavior Model: Finite State Machine

Internal validity through hard-coded **historical behavior**



Behavior Model: Q-Learning

- **Reinforcement Learning Model** - AI Model
- **Reward** of actions **influences** the agent **decisions**
- Hypothesis explores influence of **territory retention**
  - Q-learning optimizes agent **decisions**
  - Modifying Q-learning **parameters grants** agent **behaviors**



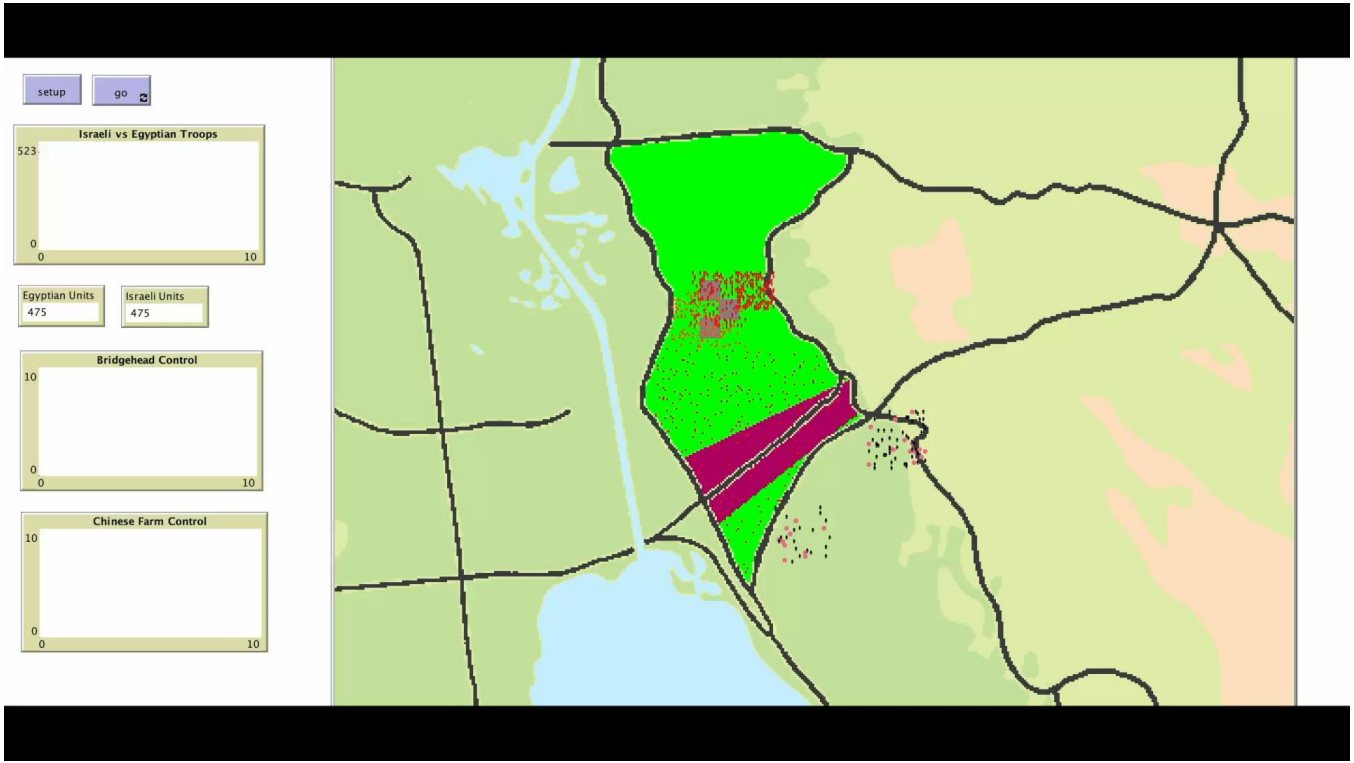
Q-Learning: Parameter Overview

Discount Factor (γ) → Tendency to **prioritize future gains** over immediate territory control

Learning Rate (α) → Rate at which agents **adapt their behavior** based on new rewards

Exploration Factor (ε) → Chance of agents trying **random action** instead of optimal decision

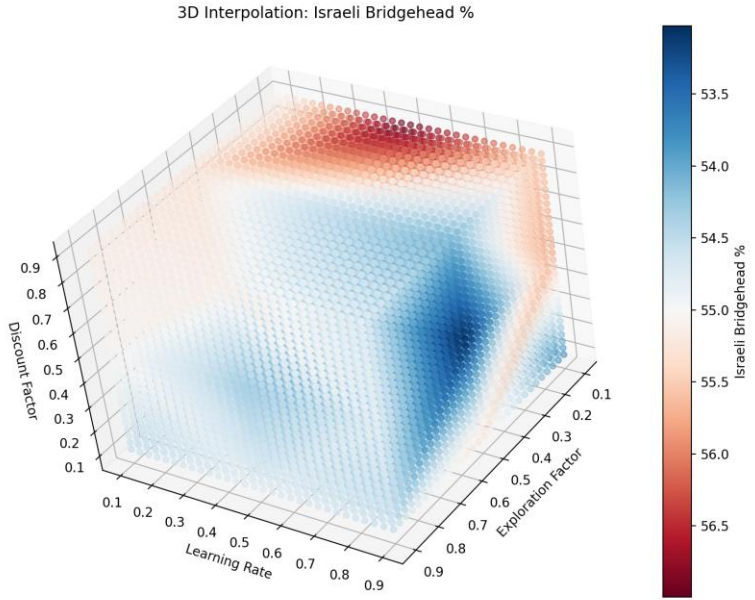
Model Demo



Key

- Chinese Farm
- Israeli Tanks
- Israeli Soldiers
- Egyptian Soldiers
- Egyptian Tanks
- Bridgehead “Zone”
- Fortified Egyptian Positions
- Landmines

Model Validation and Conditioning



160,000 data points for all combinations
1.6 minutes per run
48 hours total runtime

Internal Validity

- Interpolated all variables in a continuous format
- Range holds true battle value

External Validity

- No outliers
- No significant variance

Statistical Significance

Variable	p-value
Egyptian Bridgehead Control (%)	0.779
Israeli Bridgehead Control (%)	0.160
Chinese Farm under Egypt (%)	0.757
Chinese Farm under Israel (%)	0.340

Figure 1. Shapiro-Wilk test shows that all of the variables are normally distributed

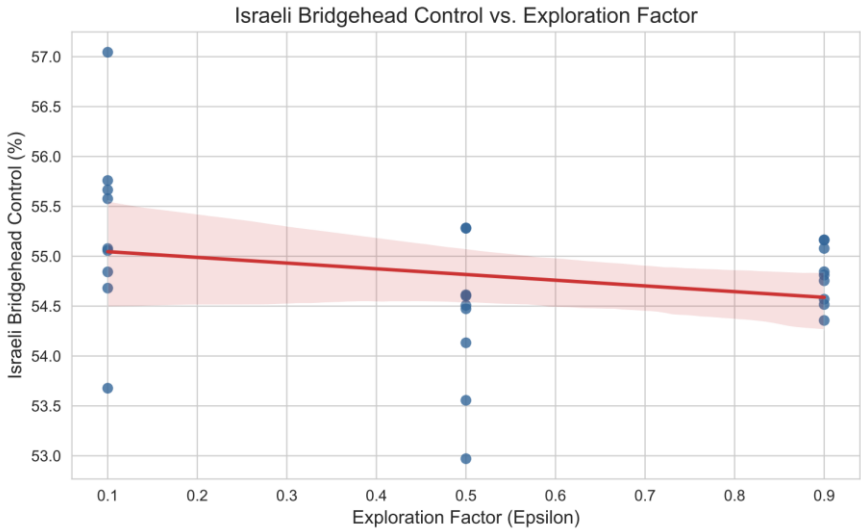
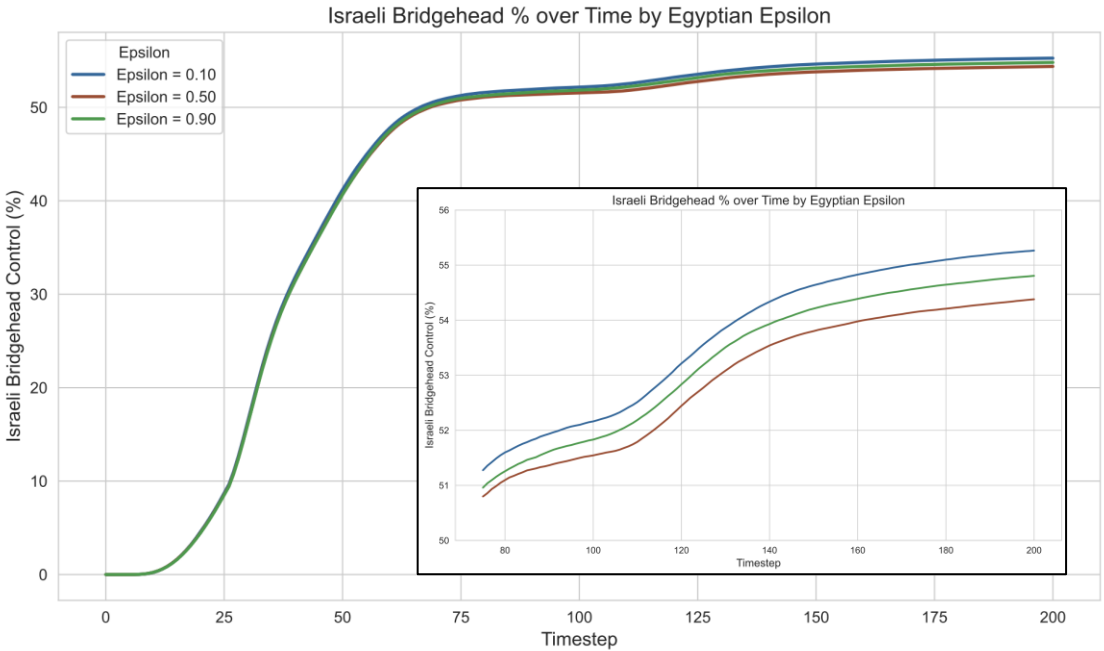


Figure 2. The exploration factor's effect on Israeli Bridgehead % Control is most statistically significant according to ANOVA test

Egyptian Exploration Factor vs Israeli Bridgehead Control %



Low Exploration Factor: Egyptian Units stick to their optimal Q-learning strategy

High Exploration Factor: Egyptian Units move to other rewards – hold other strategic positions, attack Israeli troops, etc.

Analysis

Background:

- Model framed around battle's turning point
- *RQ* - Positioning Strategy vs Territory Control

Findings:

- *Pre Timestep 75:* ε has no effect
- *Post Timestep 75:* $\varepsilon = 0.5$ optimal

Takeaways:

- Delayed penalty for static strategy
- Strategic Myopia

Model Strength - Barometer not Prescription

- Real time risk heat map: tweak vs stay the course
- Explores effects of strategic variance
- Quantifies strategic openness in decision making

Uses for a Commanding Officer

- Timing tool - long term vs. short term
- Fight false sense of stability
- Audit command style using past battles
- Identify strategic inflection points

Sources

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